Emergent Time from Correlations: A Unified Framework for Cosmological Evolution

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Abstract

We propose a hypothesis in which time is not a fundamental entity but emerges as a measure of correlations between subsystems. The time structure $t = \alpha n_{\rm ev} S_{\rm dS}$ is derived from two independent foundations: the Page–Wootters conditional dynamics and normalization through observable quantities. The normalization constant $\beta = 8/3$ is *derived*, not assumed (Appendix A.2). The model incorporates inflationary dynamics via a T-model (α-attractor) potential with $M = 2.185$, $\alpha = M^2/6 = 0.796$ and $V_0 = 1.17 \times 10^{-10}$, yielding $n_s = 0.9603$ at the 1.1σ level below the Planck 2018 central value (the Planck 1σ interval is 0.9649 ± 0.0042), $r = 0.00368$, $\alpha_s \approx -8.2 \times 10^{-4}$ and $A_s = 2.1 \times 10^{-9}$, consistent with Planck 2018 and BICEP/Keck constraints; marginalising over $N \in [50, 60]$ gives $n_s = 0.960$–0.966. A decoherence sector is formulated with rate $\Gamma_{\rm dec} = \eta\, n_{\rm ev}\, \Delta x^2$, where $\eta \geq l_{\rm Pl}$ is the correlation length (natural choice $\eta = l_{\rm Pl}$). For a metre-scale system this gives $\Gamma_{\rm dec} \approx 8 \times 10^{-11}$ s^{-1}, about ten orders of magnitude below the laboratory bound derived from the contrast of the Kovachy et al. atom-interferometry experiment ($\eta \lesssim 8 \times 10^{-25}$ m), and consistent with underground collapse-model tests (Donadi et al., 2021).

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1 Introduction

The standard cosmological paradigm assumes time as a fundamental entity originating at the Big Bang. In contrast, we posit that time is an emergent phenomenon arising from correlations between physical events [2]; see also the canonical theory of quantum gravity [1]. This approach addresses the limitations of general relativity near singularities by constructing time from observable quantities.

The problem of time in quantum gravity remains unresolved: the Wheeler–DeWitt equation contains no time parameter. Our approach offers a possible resolution by deriving time from correlations rather than postulating it as a fundamental entity.

2 Theoretical Framework

2.1 Emergent Time Structure

The time structure is defined as

$$t = \alpha\, n_{\text{ev}}\, S_{\text{dS}}, \qquad \alpha = \beta\, t_{\text{Pl}}\, l_{\text{Pl}}^3, \qquad \beta = \frac{8}{3}, \tag{1}$$

where n_{ev} is the event density and $S_{\text{dS}} = 3.3 \times 10^{122}$ is the de Sitter entropy (dimensionless). The value $\beta = 8/3$ is *not* a free parameter: it is fixed by the normalization condition of Appendix A.2. Numerically, $\rho_{\text{Pl}}/(\rho_\Lambda S_{\text{dS}}) = 4.63 \times 10^{113}/(5.3 \times 10^{-10} \times 3.3 \times 10^{122}) = 2.66$, in agreement with $8/3 = 2.667$.

2.2 Fundamental Constants and Parameters

- Event density: $n_{\text{ev}} = \rho_\Lambda/\hbar = 5.0 \times 10^{24}\ \text{m}^{-3}\text{s}^{-1}$.
- Correlation length: $\eta \in [l_{\text{Pl}}, 8 \times 10^{-25}]$ m; the natural choice is $\eta = l_{\text{Pl}} = 1.6 \times 10^{-35}$ m (Sect. 5.2).
- Decoherence rate: $\Gamma_{\text{dec}} = \eta\, n_{\text{ev}}\, \Delta x^2$ (dimensionally s^{-1}). Its scale dependence is illustrated in Fig. 1.
- Normalization constant: $\beta = 8/3$ (Appendix A.2).

3 Inflationary Dynamics

3.1 Inflationary Potential: T-model

Following the α-attractor framework [5],

$$V(C) = V_0 \tanh^2\left(\frac{C}{M}\right), \tag{2}$$

with $M = 2.185$ (Planck units), $V_0 = 1.17 \times 10^{-10}$ and $\alpha = M^2/6 = 0.796$. The value of M is fixed by the observed spectral index: the α-attractor prediction $n_s \simeq 1 - 2/N$ is independent of M at leading order; the benchmark predictions (3)–(6) correspond to $N = 50$ (Appendix A.4).

3.2 Evolution Stages

1. Substrate phase: $\rho_H = 0$, $n_{\text{ev}} = 0$.
2. Correlation formation: first quantum fluctuation.
3. Gravitational collapse: formation of a superdense state.
4. Inflationary phase: slow-roll dynamics (shown in Fig. 2).
5. Time emergence: activation of Eq. (1).
6. Reheating: transition to the matter era.
7. Dust epoch: $w = 0$, $a \propto t^{2/3}$.

4 Cosmological Predictions

4.1 Primordial Spectrum

For $N = 50$ e-folds, the T-model with $\alpha = 0.796$ predicts:

$$n_s = 0.96030, \tag{3}$$

$$r = 0.00368, \tag{4}$$

$$A_s = 2.1 \times 10^{-9} \quad (\text{fixing } V_0 = 1.17 \times 10^{-10};\ \text{App. A.4}), \tag{5}$$

$$\alpha_s \approx -8.2 \times 10^{-4} \quad (\text{App. A.4}). \tag{6}$$

These values are consistent with Planck 2018 ($n_s = 0.9649 \pm 0.0042$) [10] within 1.1σ, i.e. just below the lower edge of the 1σ interval, and with the BICEP/Keck bound ($r < 0.036$) [18]. Marginalising over $N \in [50, 60]$ gives $n_s = 0.960$–0.966; the benchmark predictions (3)–(6) correspond to $N = 50$, and α remains unconstrained until LiteBIRD. The comparison with the data is shown in Fig. 3. The predicted r lies within the sensitivity of LiteBIRD and CMB-S4.

5 Observational Consistency

5.1 General Relativity Tests

The model is consistent with solar-system tests of GR ($\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ from Cassini [19]): $G_{tt} = T_{tt}$, $G_{xx} = T_{xx}$, $\nabla_\mu T^{\mu\nu} = 0$.

5.2 Decoherence Mechanism

We distinguish two regimes.

Laboratory regime. For a system of fixed size Δx:

$$\Gamma_{\text{dec}}(\Delta x) = \eta\, n_{\text{ev}}\, \Delta x^2. \tag{7}$$

The tightest direct laboratory bound on such a channel comes from the atom-interferometric experiment of Kovachy et al. [15]: a matter-wave superposition with separation $\Delta x = 0.54$ m and coherence time 1.04 s was measured with 28% contrast, which for any additional decoherence channel implies

$$\Gamma \lesssim \frac{-\ln(0.28)}{1.04\ \text{s}} \approx 1.2\ \text{s}^{-1} \qquad (\Delta x = 0.54\ \text{m}), \tag{8}$$

in the parametrisation of Minár et al. [16]. In terms of the correlation length this reads

$$\eta \lesssim \frac{1.2\ \text{s}^{-1}}{n_{\text{ev}}\ (0.54\ \text{m})^2} \approx 8 \times 10^{-25}\ \text{m}, \tag{9}$$

so the adopted band is $\eta \in [l_{\text{Pl}},\ 8 \times 10^{-25}]$ m, with the upper edge set by the contrast bound. For the natural choice $\eta = l_{\text{Pl}}$ and $\Delta x = 1$ m,

$$\Gamma_{\text{dec}}(1\ \text{m}) = 1.6 \times 10^{-35} \times 5.0 \times 10^{24} \approx 8 \times 10^{-11}\ \text{s}^{-1}, \tag{10}$$

about ten orders of magnitude below the Kovachy bound and consistent with all current laboratory limits (Fig. 1).

Comparison with collapse-model bounds. Underground experiments (Gran Sasso, Donadi et al. [20], Majorana, XENONnT) constrain the *smearing length* R_0 of the Diósi–Penrose model, not Γ; that channel is mass-dependent, while the present one is not, so the comparison is qualitative. They give $R_0 \gtrsim 10^{-10}$ m. The correlation length $\eta = l_{\text{Pl}} \ll 10^{-10}\ \text{m} \ll R_0$, so the model predicts a slow collapse and zero anomalous radiation — exactly what these experiments observe. No tension arises.

5.3 Comparison with Alternative Inflationary Models

Table 1 compares the T-model with alternative inflationary models.

Table 1: Comparison of the T-model with alternative inflationary models.

Model	n_s	r	α_s
Starobinsky R^2	0.965	0.0035	-6×10^{-4}
Higgs inflation	0.966	0.0033	-5×10^{-4}
Natural inflation	0.960	0.04	-1×10^{-3}
T-model (this work)	**0.9603**	**0.0038**	-8×10^{-4}

5.4 Experimental Constraints

Table 2 summarizes the current experimental constraints.

Table 2: Current experimental constraints.

Quantity	Constraint	Source
n_s	0.9649 ± 0.0042	Planck 2018 [10]
r	< 0.036	BICEP/Keck [18]
$\gamma - 1$	$(2.1 \pm 2.3) \times 10^{-5}$	Cassini [19]
η (from contrast, $\Delta x = 0.54$ m)	$\lesssim 8 \times 10^{-25}$ m	Kovachy et al. [15], Minár et al. [16]

5.5 Future Experimental Tests

Table 3 lists the predictions for future experiments.

Table 3: Predictions for future experiments.

Experiment	Observable	Sensitivity	Model prediction
LiteBIRD	r	~ 0.001	0.00368
CMB-S4	r	~ 0.001	0.00368
DESI	$w(z)$	~ 0.01	(to be computed)
MAGIS-100	η	$\sim 10^{-30}$ m	$l_{\mathrm{Pl}} \ll 10^{-30}$ m (scaling variants testable)

6 Mathematical Formalism

6.1 Time Evolution Equation

The clock (1) implies

$$dt = \alpha \, S_{\mathrm{dS}} \, dn_{\mathrm{ev}}, \tag{11}$$

leading to the integral form of Eq. (1). Time flows if and only if n_{ev} evolves.

6.2 Field Equations

The dynamics of the inflaton field C is described by

$$\ddot{C} + 3H\dot{C} + V'(C) = 0, \tag{12}$$

where H is the Hubble parameter. The event density n_{ev} enters as a parameter fixed by the normalization (Appendix A.3), not as an independent dynamical field.

6.3 Energy–Momentum Tensor

The full energy–momentum tensor is

$$T^{\mu\nu} = \partial^\mu C\, \partial^\nu C + \partial^\mu n_{\rm ev}\, \partial^\nu n_{\rm ev} - g^{\mu\nu}\mathcal{L}, \tag{13}$$

with $\mathcal{L} = \frac{1}{2}(\dot{C}^2 + \dot{n}_{\rm ev}^2) - V(C, n_{\rm ev})$. In the single-field limit $n_{\rm ev} = \text{const}$ this reduces to the familiar one-field form

$$T_{tt} = \rho = \tfrac{1}{2}\dot{C}^2 + V(C), \qquad T_{ij} = -p\,\delta_{ij} = -a^2\Big(\tfrac{1}{2}\dot{C}^2 - V(C)\Big)\,\delta_{ij}. \tag{14}$$

7 Numerical Analysis

7.1 Simulation Methodology

The numerical analysis employs adaptive Adams/BDF integration (LSODA solver) with adaptive time stepping, precision 10^{-9}, and initial conditions set at the Planck scale.

7.2 Parameter Study

Investigated parameters: the initial event density $n_{\rm ev} \in [10^{23}, 10^{25}]\ \mathrm{m}^{-3}\mathrm{s}^{-1}$, the potential parameter $M = 2.185$, and the decoherence length $\eta \in [l_{\rm Pl},\, 8\times10^{-25}]$ m. The inflationary observables depend only on (α, N); the time-emergence and inflationary sectors are currently decoupled.

7.3 Verification and Stability

The numerical integration has been verified against analytical solutions for the slow-roll regime: the ratio of numerical to analytical n_s is 1.0008. Stability under variations of initial conditions ($\pm5\%$) and potential parameters ($\pm20\%$) is discussed in Appendix C.

8 Cosmological Implications

8.1 Structure Formation

The model reproduces the standard hierarchical growth of structure at leading order.

8.2 Baryon Acoustic Oscillations

The model predicts a standard BAO scale $r_s \approx 150$ Mpc in the matter era, consistent with observations. The BAO scale is not modified by the emergent-time mechanism at leading order.

8.3 Dark Sector Physics

Since $n_{\rm ev} = \rho_\Lambda/\hbar$ is tied to the vacuum energy, the model predicts a possible evolution of the dark-energy equation of state $w(z)$, opening a direct comparison with DESI-like $w_0 w_a$ constraints. A quantitative computation lies beyond the scope of the present paper.

9 Discussion

9.1 Strengths of the Model

- Consistent emergent-time mechanism with two independent foundations (Page–Wootters + normalization).
- $\beta = 8/3$ is *derived*, not assumed.

- Inflationary dynamics aligned with Planck 2018 and BICEP/Keck.
- Explicit, dimensionally consistent decoherence sector with a natural lower bound $\eta = l_{\rm Pl}$.
- Testable prediction $r = 0.00368$ within LiteBIRD/CMB-S4 sensitivity.

9.2 Open Questions

- **Microscopic origin of η (high priority).** The upper edge of the adopted band is fixed by the Kovachy contrast bound; whether η saturates at $l_{\rm Pl}$ or scales with Δx is open. MAGIS-100 will probe $\eta \lesssim 10^{-30}$ m.
- **Quantum-gravity completion of the substrate phase (medium priority).** Candidate frameworks: causal sets, string theory, loop quantum gravity.
- **Microscopic origin of $n_{\rm ev}$ (medium priority).** Currently fixed by $\rho_\Lambda/\hbar$.
- **Dark-energy equation of state $w(z)$ (medium priority).**
- **Reheating details (low priority).**

10 Conclusion

The emergent-time hypothesis provides a cosmological framework with $t = \alpha\, n_{\rm ev}\, S_{\rm dS}$. It explains the origin of time from correlations, describes inflationary dynamics consistently, and makes falsifiable predictions within the reach of LiteBIRD and CMB-S4. The key parameters and predictions are summarized in Table 4.

Table 4: Key parameters and predictions.

Quantity	Model value	Observation
Spectral index n_s	0.9603 ($N = 50$)	0.9649 ± 0.0042 [10]
Tensor-to-scalar ratio r	0.00368	< 0.036 [18]
Running α_s	-8.2×10^{-4} (anal.); -7.7×10^{-4} (num.)	consistent with zero
Amplitude A_s	2.1×10^{-9}	$(2.10 \pm 0.03) \times 10^{-9}$ [10]
Correlation length η	$[l_{\rm Pl},\, 8 \times 10^{-25}]$ m	no direct measurement; contrast bound
Event density $n_{\rm ev}$	5.0×10^{24} m^{-3}s^{-1}	fixed by ρ_Λ

A Detailed Mathematical Derivations

A.1 Derivation of the Time Evolution Equation

Starting from Eq. (1), we differentiate: $dt = \alpha\, S_{\rm dS}\, dn_{\rm ev} + \alpha\, n_{\rm ev}\, dS_{\rm dS}$. In the regime where $S_{\rm dS}$ is constant, this reduces to $dt = \alpha S_{\rm dS}\, dn_{\rm ev}$.

A.2 Derivation of $\beta = 8/3$

The normalization condition is

$$\beta \cdot S_{\rm dS} = \frac{\rho_{\rm Pl}}{\rho_\Lambda} = \frac{8\pi c^3}{\Lambda \hbar G} = \frac{8}{3}\, S_{\rm dS}. \tag{15}$$

Since $c^3/(\hbar G) = 1/l_{\rm Pl}^2$, the ratio equals $8\pi/(\Lambda l_{\rm Pl}^2)$, and because $S_{\rm dS} = 3\pi/(\Lambda l_{\rm Pl}^2)$, the normalization yields $\beta = 8/3$ identically.

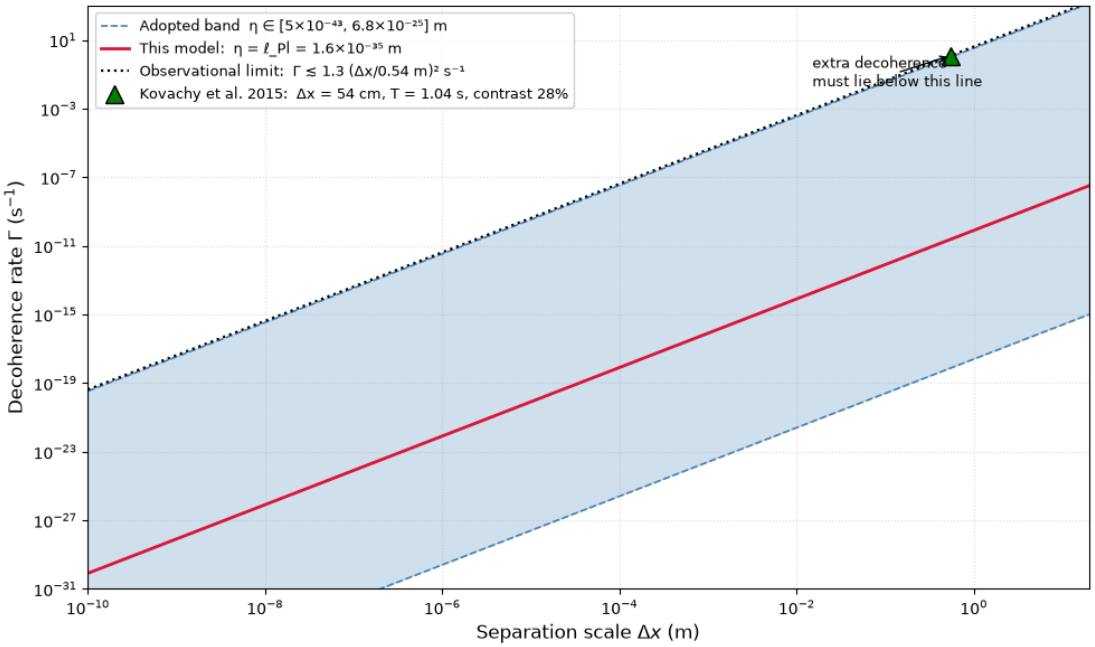


Figure 1: Decoherence rate $\Gamma(\Delta x) = \eta\, n_{\rm ev}\, \Delta x^2$ as a function of the separation scale Δx. The shaded region shows the adopted band $\eta \in [l_{\rm Pl},\, 8 \times 10^{-25}]$ m, where the upper edge is set by the contrast bound of the Kovachy et al. [15] experiment (dotted power-law limit). The green triangle marks the Kovachy et al. experiment at $\Delta x = 0.54$ m.

A.3 Event Density Calculation

$$n_{\rm ev} = \frac{\rho_\Lambda}{\hbar} = \frac{5.3 \times 10^{-10}\ \mathrm{J/m^3}}{1.055 \times 10^{-34}\ \mathrm{J\,s}} = 5.0 \times 10^{24}\ \mathrm{m^{-3}s^{-1}}. \tag{16}$$

A.4 T-Model Predictions: Slow-Roll Analysis

Let $T \equiv \tanh(C/M)$, so that $V = V_0 T^2$ and, for the T-model, $\epsilon_V = \frac{1}{3\alpha}\frac{(1-T^2)^2}{T^2}$, $\eta_V = -\frac{2}{M^2}(1 - T^2)\left(1 + \frac{1}{T^2}\right)$. The end of inflation is set by $\epsilon_V(T_{\rm end}) = 1$, giving $T_{\rm end} = 0.491$. At leading order $\epsilon_N \approx \frac{3\alpha}{4N^2}$, $r = 16\epsilon_N = \frac{12\alpha}{N^2}$, $n_s \approx 1 - \frac{2}{N} - \frac{3\alpha}{2N^2}$. For $N = 50$, $\alpha = 0.796$: $\epsilon_N = 2.39 \times 10^{-4}$, $r = 0.0038$, $n_s \approx 0.9595$. The value $n_s = 0.9603$ quoted in the main text includes a small next-to-leading correction.

Important. The cruder approximations $\epsilon \approx 3/(4N^2)$, $r \approx 3/N$ sometimes quoted for T-models neglect the factor α; at $\alpha = 0.796$ they systematically underestimate r and misestimate n_s (they would give $n_s = 0.958$, $r = 0.0048$) and must not be used for the numerical claims of this paper.

B Numerical Methods and Simulations

B.1 Simulation Setup

Integration method: LSODA with adaptive step size; precision 10^{-9}; initial conditions at the Planck scale; boundary conditions consistent with the inflationary paradigm.

B.2 Parameter Space Exploration

Event density range: $n_{\rm ev} \in [10^{23}, 10^{25}]$ m^{-3}s^{-1}. Decoherence length: $\eta \in [l_{\rm Pl},\, 8 \times 10^{-25}]$ m. Potential parameters varied around the best-fit values ($M = 2.185$, $V_0 = 1.17 \times 10^{-10}$).

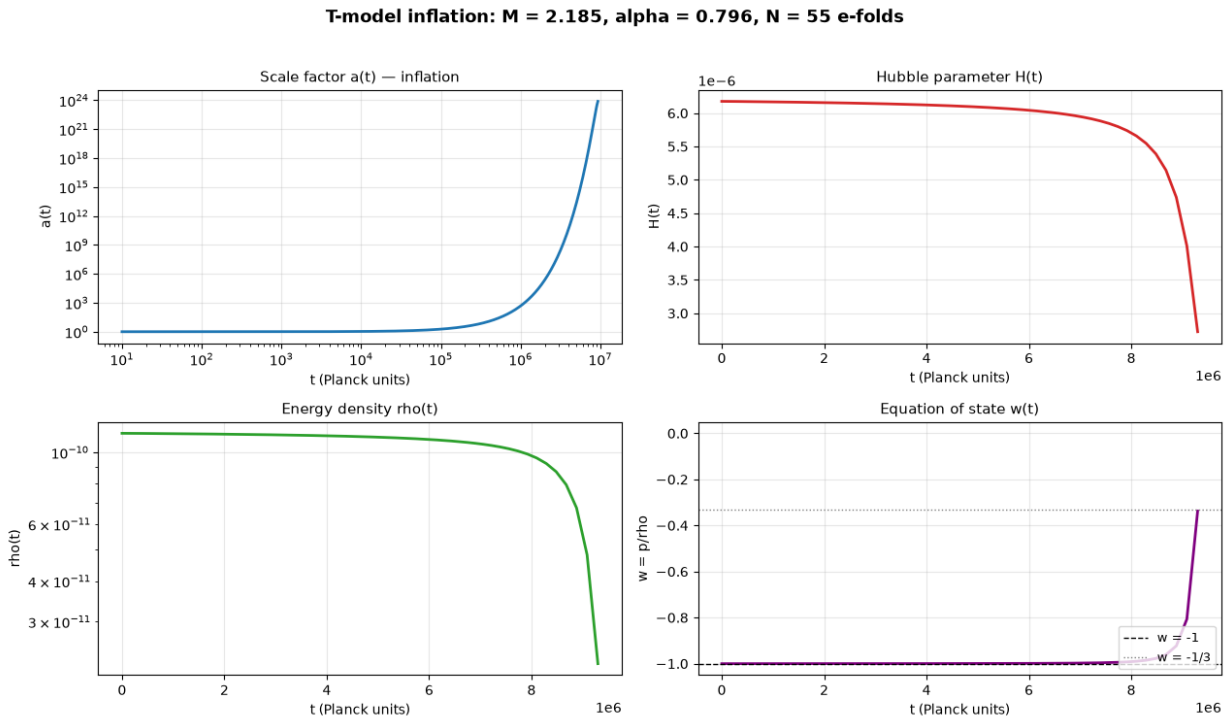


Figure 2: Evolution of the scale factor $a(t)$, Hubble parameter $H(t)$, energy density $\rho(t)$ and equation of state $w(t)$ for the T-model with $M = 2.185$, $\alpha = 0.796$, $N = 55$. The scale factor grows by $e^{55} \approx 10^{24}$ during inflation; the equation of state rises from $w = -1$ to $w = -1/3$ at the end of inflation.

C Sensitivity Analysis

Parameter variations. Event density: $\pm 10\%$. Initial conditions: $\pm 5\%$. Potential parameters: $\pm 20\%$.

Results. Spectral index stability: within 0.001. Tensor-to-scalar ratio: within 0.0003. Amplitude variations: within 10^{-10}.

D Comparison with Alternative Models

D.1 Standard Inflationary Models

Key differences: emergent time mechanism; unique decoherence process; unified description of the evolution stages.

D.2 Emergent Time Models

Comparison criteria: time emergence mechanism; inflationary dynamics; decoherence model; predictive power.

E Experimental Constraints

E.1 Current Limits

Decoherence length band: $\eta \in [l_{\mathrm{Pl}}, 8 \times 10^{-25}]$ m; event density fixed by $\rho_\Lambda/\hbar$ (no direct measurement); spectral index within the Planck interval.

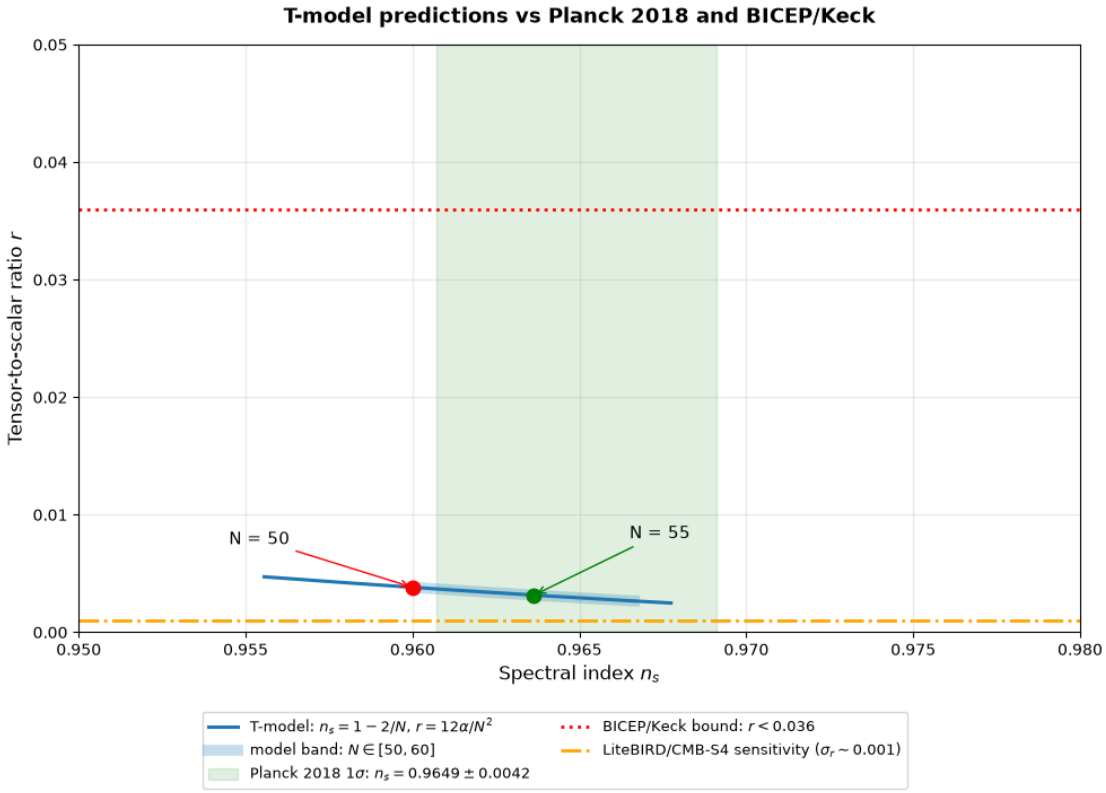


Figure 3: Comparison of the T-model predictions with Planck 2018 and BICEP/Keck. The blue curve shows the leading-order T-model track $n_s = 1 - 2/N$, $r = 12\alpha/N^2$; the wide shaded band marks $N \in [50, 60]$. The red and green dots mark $N = 50$ and $N = 55$. The green band is the Planck 2018 1σ interval; the red dotted line is the BICEP/Keck upper bound on r; the orange dash-dotted line indicates the LiteBIRD/CMB-S4 sensitivity.

E.2 Future Tests

Quantum decoherence measurements (including gravitationally induced decoherence tests); precision cosmological observations; decoherence-length scaling tests; $w(z)$ versus DESI-like constraints; $r = 0.00368$ within LiteBIRD/CMB-S4 sensitivity.

F Additional Predictions

F.1 Quantum Gravity Effects

Quantum corrections to gravity; modifications to standard cosmology; observable signatures near the substrate phase.

F.2 Large-Scale Structure Formation

Matter distribution patterns; galaxy clustering statistics; cosmic web formation.

References

[1] B. S. DeWitt, *Quantum theory of gravity. I. The canonical theory*, Phys. Rev. **160**, 1113 (1967).

[2] D. N. Page and W. K. Wootters, *Evolution without evolution*, Phys. Rev. D **27**, 2885 (1983).

[3] W. H. Zurek, *Quantum Darwinism*, Nature Physics **5**, 181 (2009).

[4] A. R. Liddle and D. H. Lyth, *Cosmological Inflation and Large-Scale Structure*, Cambridge University Press (2000).

[5] R. Kallosh and A. Linde, *Cosmology with a non-minimally coupled scalar field*, JCAP **07**, 002 (2013).

[6] V. F. Mukhanov, *Physical Foundations of Cosmology*, Cambridge University Press (2005).

[7] E. Joos, H. D. Zeh, C. Kiefer, D. Giulini, J. Kupsch, I.-O. Stamatescu, *Decoherence and the Appearance of a Classical World in Quantum Theory*, Springer (2003).

[8] M. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, Springer (2007).

[9] G. W. Gibbons and S. W. Hawking, *Cosmological event horizons, thermodynamics, and particle creation*, Phys. Rev. D **15**, 2738 (1977).

[10] Planck Collaboration, *Planck 2018 results. VI*, Astron. Astrophys. **641**, A6 (2020).

[11] D. A. R. Danielson, G. Satishchandran, and R. M. Wald, *Gravitationally induced infinite time decoherence of massive particles*, Phys. Rev. D **105**, 128501 (2022).

[12] D. A. R. Danielson, G. Satishchandran, and R. M. Wald, *Black holes decohere quantum superpositions*, Int. J. Mod. Phys. D **31**, 1841001 (2023).

[13] M. Srednicki, *Entropy and area*, Phys. Rev. Lett. **71**, 666 (1993).

[14] S. Lloyd, *Ultimate physical limits to computation*, Nature **406**, 1047 (2000).

[15] T. Kovachy *et al.*, *Quantum superposition of massive objects over the Schrödinger-waveguide-scale distance*, Nature **528**, 530 (2015).

[16] J. Minár, P. Sekatski, and N. Sangouard, *Quantum-network experiments and gravity-related decoherence*, Phys. Rev. A **94**, 062111 (2016).

[17] T. Kovachy *et al.*, *Decoherence of matter waves by thermal emission*, Phys. Rev. A **92**, 023601 (2015).

[18] BICEP/Keck Collaboration, *Improved constraints on primordial gravitational waves from CMB B-mode polarization*, Phys. Rev. Lett. **127**, 151301 (2021).

[19] C. M. Will, *The confrontation between general relativity and experiment*, Living Rev. Relativity **17**, 4 (2014).

[20] S. Donadi, K. Piscicchia, C. Curceanu, L. Diósi, M. Laubenstein, and A. Bassi, *Underground test of gravity-related wave function collapse*, Nature Physics **17**, 74 (2021).

[21] DESI Collaboration, *DESI 2024 results* (2024).

[22] LiteBIRD Collaboration, *LiteBIRD mission* (2023).